MIL-C-24643/7B
3 October 1986
SUPERSEDING
MIL-C-0024643/7A(SH)
8 November 1985
MIL-C-24643/7
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSMHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - AWG size 16, class K stranding, uncoated.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation, nominal wall thickness 0.020 inch. Standard identification code applied by method 1.
Fourth - The specified number of conductors (see table I) cabled together in one or more layers, all in left-hand direction, with the minimum amount of fillers necessary to form a firm, round assembly. Cabling sequence shall be consecutive, starting with no. 1, from center outward. The length of lay shall be determined to permit the completed cable to meet the twisting and bending endurance tests. A single conductor shall not be used as center for the cable core.
Fifth - Binder tape applied helically with overlap.
Sixth - Cross-linked polyolefin jacket. Nominal wall thickness 0.060 inch. Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
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TABLE I. Details.

Military part no. M24643/7	Type and size	Number of conductors	Overall diameter		Bending endurance roller diameter (maximum) (inches)
			minimum (inch)	maximum (inch)	
-01UN	LSMHOF-7	7	0.465	0.500	2.50
-02UN	LSMHOF-10	10	.540	.585	2.50
-03UN	LSMHOF-14	14	.585	.635	3.00
-04UN	LSMHOF-19	19	.650	.705	3.00
-05UN	LSMHOF-24	24	.735	.795	3.50
-06UN	LSMHOF-30	30	.775	.835	4.00
-07UN	LSMHOF-37	37	.855	.925	4.00
-08UN	LSMHOF-44	44	.925	1.000	4.50
-09UN	LSMHOF-61	61	1.100	1.175	5.00

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum..	4.59
Voltage withstand - volts, root mean square, minimum....	2000
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.....	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Twisting and bending endurance (see table I for roller diameter)	
At 75°C, cycles, minimum.....	6000
At minus 30 ± 2°C, cycles, minimum.....	2000

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Permanence of printing (conductor) - cycles, minimum	50
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)(125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
7 and 10	1500
14 through 61	1500

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-07)

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